

Flipkart E-Commerce Sales, Customer & Return Analytics

Tools Used: Excel, Power Query, Power BI | Project Duration: 2 Weeks

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1. Project Introduction

Flipkart, India's leading e-commerce marketplace, manages vast amounts of customer, product, sales, and return data. This project analyzes sales performance, customer purchasing behavior, product profitability, and return trends. By integrating multiple datasets and applying systematic cleaning and visualization, the project delivers actionable insights to optimize revenue growth, customer retention, product strategy, and operational efficiency.

2. Problem Statement

- Sales trends and profitability are unclear due to inconsistent raw data.
- High return rates reduce overall profit margins.
- Customer purchasing behavior and segment value are not fully understood.
- Management lacks visibility into product performance and operational inefficiencies.

3. Business Objectives

- Evaluate sales, orders, quantity, profitability, and profit margin.
- Identify high-performing and underperforming products, categories, and brands.
- Understand customer purchasing behavior and segment contribution.
- Analyze regional and return performance.
- Build an interactive Power BI dashboard to convert cleaned data into actionable insights.

4. Dataset Information

Dataset Name: Flipkart Retail & E-Commerce Data

Format: CSV | Tools Used: Excel, Power Query, Power BI

Source Files	Records	Owner	Refresh
Orders.csv	3,000	Data Analytics Team	Monthly
Customers.csv	500	Data Analytics Team	Monthly
Products.csv	50	Data Analytics Team	Monthly
Returns.csv	420	Data Analytics Team	Monthly

5. Data Understanding

Dataset profiling was carried out to understand structure, completeness, and relationships.

Column	Description	Data Type	Example	Issues
Order_ID	Unique order identifier	Text	010001	None

Order_Date	Date when order was placed	Date	15/06/2025	64 inconsistent formats
Customer_ID	Customer reference	Text	C001	73 missing values
Product_ID	Product reference	Text	P001	None
Quantity	Units ordered	Whole Number	2	19 invalid (zero/negative)
Sales	Revenue generated	Decimal	₹24,999.50	118 blanks
Profit	Profit generated	Decimal	₹6,999.50	Some negative values
Customer_Segment	Customer classification	Text	Consumer	Missing in some rows
State	Customer/order state	Text	Telangana	38 mismatches with City
City	Customer/order city	Text	Hyderabad	Geographic inconsistencies
Category	Product category	Text	Electronics	None
Sub_Category	Product sub-category	Text	Laptop	None
Brand	Product brand	Text	Dell	Inconsistent case/spelling
Return_Reason	Reason for return	Text	Damaged	Missing values
Refund_Amount	Amount refunded	Decimal	₹2,500.00	17 missing values

Data Quality Report Template

Customer.csv

Issue	Column	Count	Action Required
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Missing Values	Customer_Name	16	Fill missing names from source
Missing Values	City	14	Update missing city values from source
Missing Values	Customer_ID	0	No action required
Missing Values	Customer_Segment	0	No action required
Missing Values	Gender	0	No action required
Missing Values	Age_Group	0	No action required
Missing Values	Region	0	No action required
Missing Values	State	0	No action required

6. Data Cleaning

Step	Transformation	Reason	Completed
1	Imported Orders, Customers, Products, Returns	Unified environment for cleaning	Yes
2	Renamed queries and standardized column names	Consistency across datasets	Yes
3	Changed data types for IDs, dates, numeric fields	Accurate calculations	Yes
4	Trimmed and cleaned text fields	Removed spaces and hidden characters	Yes
5	Standardized case, spelling, category values	Uniformity in categories/brands	Yes
6	Handled null and blank values	Applied field-specific rules	Yes
7	Removed duplicate records	Prevented distortion in analysis	Yes
8	Standardized mixed	Reliable time-series	Yes

	date formats	analysis	
9	Validated numeric fields (Quantity, Sales, Profit, Discount, Refund)	Corrected invalid/zero/extreme values	Yes
10	Reviewed abnormal Profit values	Differentiated genuine losses vs errors	Yes
11	Validated State-City combinations	Corrected geographic inconsistencies	Yes
12	Verified Customer_ID and Product_ID references	Ensured relational integrity	Yes
13	Validated Order_ID references in Returns	Linked returns to valid orders	Yes
14	Created calculated columns (Profit Status)	Classified records for analysis	Yes
15	Final quality checks	Confirmed datasets ready for modeling	Yes

7. Exploratory Data Analysis

Analysis Area	Exploration Performed	Key Observations
Sales Distribution	Category-wise and product-level	Fashion highest sales (₹5.28 Cr)
Profitability	Category and product margins	Electronics stronger margins than Fashion
Product Insights	Top-selling and low-margin products	Hair Dryer Max top sales; Face Wash Classic low margin
Customer Segmentation	Segment contribution	Consumer segment ₹8.78 Cr sales, ₹94.27 Lakh profit
Regional Analysis	State and city-level	Hyderabad 297 returns, top hotspot

Return Trends	Product and category returns	Skin Care Kit X & Sneakers 100 highest returns (13 each)
Temporal Trends	Monthly sales	Seasonal peaks and dips observed

8. Visualizations Used

KPI Cards:

Visualization	Purpose
Total Sales	Tracks overall revenue generated
Total Profit	Measures profitability across transactions
Order Count	Indicates total number of orders placed
Customer Count	Shows active customer base size
Profit Margin %	Evaluates efficiency of sales relative to profit
Average Order Value	Assesses customer spending per order
Return Count	Total number of returned orders
Return Rate %	Percentage of orders returned
Total Refund	Monetary value refunded to customers
Profit After Refund	Net profit after refunds deducted
Loss Making Orders	Identifies transactions with negative profitability
Average Refund	Average refund amount per return

Charts:

Visualization	Purpose
Top 10 Products	Highlights best-selling products
Sales & Profit by Category	Compares revenue and profitability across categories
Sales by State	Regional performance analysis

Monthly Sales Trend	Tracks revenue fluctuations over time
Customer Segment Performance	Evaluates contribution of Consumer, Corporate, Home Office segments
Brand Performance	Identifies top-performing brands
Return by Category	Shows categories with highest return counts
Return Reasons	Analyzes customer-reported reasons for returns
Profit vs Refund	Compares profitability against refund amounts
Return Hotspots	Identifies cities/states with highest return activity
Segment Return Analysis	Evaluates return behavior across customer segments
Transaction Details (Table)	Provides granular view of return transactions

9. Dashboard Design

Dashboard 1 – Sales & Customer Performance

KPI Cards:

- Total Sales
- Total Profit
- Order Count
- Customer Count
- Profit Margin %
- Average Order Value

Visuals:

- Top 10 Products
- Sales & Profit by Category
- Sales by State
- Monthly Sales Trend
- Customer Segment Performance

- Brand Performance

Dashboard 2 – Profitability & Returns Analysis

KPI Cards:

- Return Count
- Return Rate %
- Total Refund
- Profit After Refund
- Loss Making Orders
- Average Refund

Visuals:

- Return Analysis by Category
- Return Reason Distribution
- Refund vs Profit by Category
- Top Returned Products
- State-wise Return Activity & Refund Impact
- Detailed Transaction Table

10. Dashboard Screenshots

Dashboard 1: Sales & Customer Analysis Dashboard

Dashboard 2: Returns & Refunds Analytics Dashboard



Figure 2: Returns & Refunds Analytics Dashboard Overview

Figure 1: Sales & Customer Analysis Dashboard Overview

Dashboard 2: Profitability & Returns Analysis Dashboard

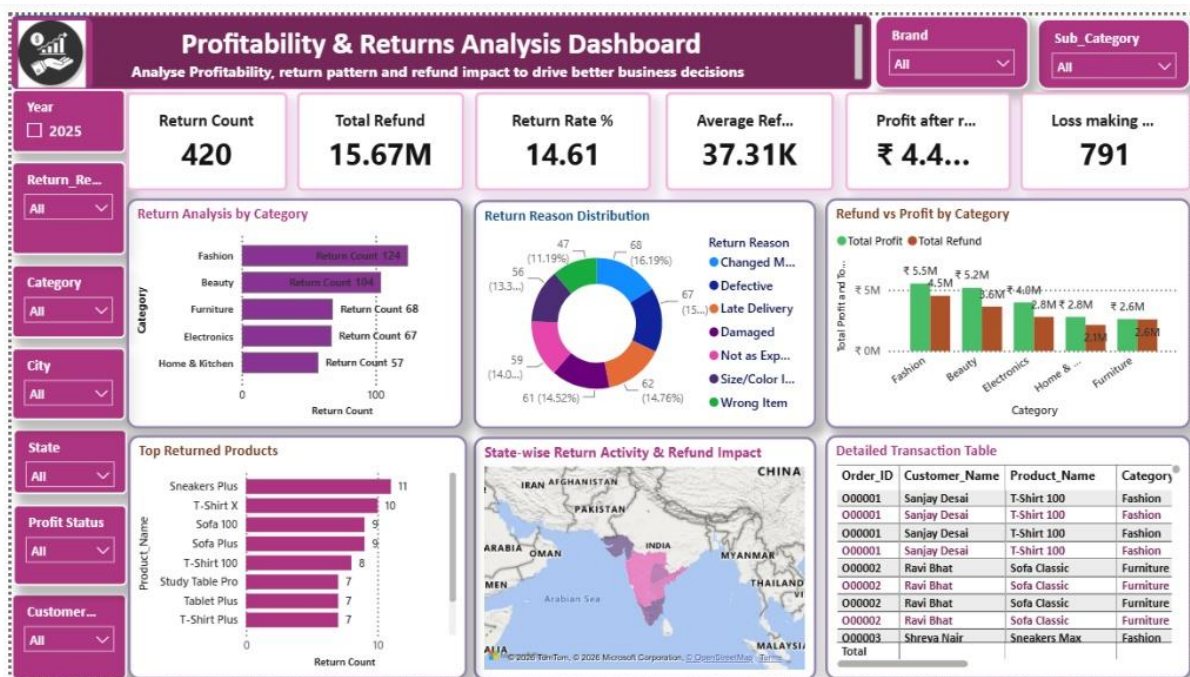


Figure 2: Profitability & Returns Analysis Dashboard Overview

11. Key Findings

- Fashion category generated the highest revenue (₹5.28 Cr).
- Electronics delivered stronger profit margins despite lower sales.
- Hair Dryer Max was the top-selling product (₹55.05 Lakh).
- Face Wash Classic had high sales but weak profitability (₹3.73 Lakh profit).
- Consumer segment dominated with ₹8.78 Cr sales and ₹94.27 Lakh profit.
- Hyderabad recorded 297 returns, making it the top return hotspot.
- Skin Care Kit X and Sneakers 100 had the highest return counts (13 each).

12. Business Insights

- Protect strongest revenue categories (Fashion & Beauty).
- Investigate high-sales, low-profit products (Face Wash Classic).
- Focus on Hyderabad returns (297 cases).
- Review frequently returned products (Skin Care Kit X, Sneakers 100).
- Prioritize Consumer segment (₹8.78 Cr sales).
- Balance sales with profitability (Electronics shows stronger margins).

13. Recommendations

- Optimize discounting and pricing strategies for low-margin products.
- Improve product quality and specifications for frequently returned items.
- Conduct deeper return-reason analysis in Hyderabad.
- Strengthen loyalty programs and personalized offers for Consumer segment.
- Balance promotional focus between high-sales and high-profit categories.
- Enhance fulfillment and customer experience to reduce returns.

14. Business Impact

- Fashion category drove ₹5.28 Cr in sales, confirming it as the strongest revenue contributor.
- Electronics delivered higher profit margins, underscoring the need to balance sales with profitability.
- Hair Dryer Max achieved ₹55.05 Lakh sales, supporting inventory prioritization for high-demand products.

- Face Wash Classic generated ₹45.97 Lakh sales but only ₹3.73 Lakh profit, highlighting margin optimization needs.
- Consumer segment contributed ₹8.78 Cr sales and ₹94.27 Lakh profit, making it the most valuable customer group.
- Hyderabad recorded 297 returns, requiring targeted return-reduction strategies.
- Frequent return products (Skin Care Kit X, Sneakers 100) provide clear focus areas for quality improvement.

15. Challenges Faced

- Inconsistent data formats across datasets.
- Duplicate records in Orders dataset.
- Missing values in Customer_ID and Refund Amount fields.
- Incorrect data types requiring validation.
- Invalid numeric entries (negative/zero values).
- Geographic mismatches between State and City fields.
- Dashboard performance optimization due to large datasets and complex calculations.

16. Conclusion

The Flipkart E-Commerce Sales, Customer & Return Analysis project successfully transformed raw datasets into actionable insights using Excel, Power Query, and Power BI. The analysis highlighted Fashion as the leading sales category, Hair Dryer Max as the top-selling product, and Consumer as the most valuable customer segment. At the same time, return analysis revealed Hyderabad as a critical hotspot and identified products with high return rates. Overall, the project provides management with a comprehensive view of sales, profitability, customer behavior, and returns, enabling better decision-making and improved business outcomes.